

Table 7. Comparison of representative LLM-based cyber-attack methods in mobile-infrastructure networks.

Ref.	Agent Framework / Example	Network Type	Primary Attack Vector
[6, 54, 55, 67, 169]	AttackLLM multi-agent pentester; LLMPot industrial honeypot; ChatIoT on-device assistant	Constrained edge / IIoT gateways	Automated scanning, firmware takeover, process hijack
[5, 85]	PLLM-CS telemetry analyser; LEO-SDN LLM-aided routing monitor	LEO constellation & ground segment	Telemetry spoofing, routing manipulation
[3, 12, 129]	Generative-replay IDS; compact-Transformer router monitor	Dynamic MANET / VANET clusters	Sybil node injection, route disruption
[30, 156, 168, 179, 186]	GenAI CAN-log anomaly detector; HackerGPT for automated exploitation; fine-tuned GPTs for CAN fuzzing; polymorphic malware generators bypassing rule-based gateways	6G-V2X communication links; in-vehicle CAN buses; ADAS sensors (e.g., LiDAR, GPS)	CAN message fuzzing to disable controls; sensor spoofing (e.g., fake GPS or LiDAR input to trigger emergency braking); SYN flood attacks
[106, 151, 166]	Net-GPT MITM for forged C2; Bayesian/LSTM hybrid IDS	UAV C2 links	Command hijack, GPS spoof, jamming
[2, 20, 99]	GPT-augmented anomaly IDS; ChatGPT-based toolkits	Acoustic & optical UWNs	Adaptive DoS floods, topology inference

proof-of-concept exploits, e.g., photon-number-splitting or detector-blinding scripts, against BB84 and decoy-state systems in real time [9]. In the future, attacks might target quantum repeaters, entanglement distribution systems, or even quantum routers, as full quantum networks develop.

4.7 Lessons Learned for Blue Teams

- (1) **Use AI to Counter AI Threats:** Deploy LLM-based monitoring systems to detect and respond to attacks from LLM-based agents. This is particularly important for complex environments like 6G networks, where defensive LLMs can identify subtle malicious patterns that humans might miss in regular operations.
- (2) **Implement Zero Trust Architecture:** In environments where LLM-based agents can automate reconnaissance and lateral movement, blue teams need to adopt zero-trust approaches that can continuously verify all users and actions, implement strict network segmentation, and never assume internal traffic is automatically trustworthy.

5 Cyberattack Capabilities of LLMs-based Agents on Mobile Infrastructure Networks

To systematically examine the threat landscape in mobile-infrastructure networks, we categorize representative scenarios according to their underlying network architectures, mobility patterns, and security challenges. In mobile infrastructure networks, LLM-based agents succeed by continually re-planning in response to wireless volatility and connectivity changes. Through tool-chaining, an agent processes telemetry, GNSS, spectrum, and LiDAR data to compose protocol-aware payloads that adjust channels in real time. This capability enables GNSS spoofing, MitM, and DDoS attacks, reducing time-to-impact from hours to milliseconds. We summarize LLM-based cyberattack capabilities across six mobile infrastructure network categories as shown in Table 7.

5.1 Internet of Things

The Internet of Things (IoT) often has constrained devices, and LLM-based agents might seek out weak links like unpatched IoT firmware or default credentials to take over devices in the IoT Supply Chain [54, 55, 169]. Ferrag *et al.* in [67] demonstrate that LLMs integrated with RAG pipelines effectively process heterogeneous telemetry and derive

threat indicators autonomously, reducing reconnaissance costs for potential attackers. The AttackLLM [6] implements an LLM-based multi-agent system for industrial attacks, outperforming human experts in water-treatment plant testing.

In vulnerability discovery, LLMs demonstrate significant capabilities. Binhulayil *et al.* successfully fine-tune a distilled model using CVE descriptions, achieving state-of-the-art F_1 scores in identifying buffer-overflow and injection vulnerabilities within embedded firmware [36]. On the defensive front, Vasilatos *et al.* present LLMPOT, an innovative LLM-controlled honeypot that implements industrial protocols and simulates physical processes, effectively attracting autonomous adversaries while identifying their LLM signatures [187]. The integration of conversational agents within constrained devices represents an emerging trend. ChatIoT enables the transformation of open-weight models into on-device security assistants capable of managing scanning, patch generation, and real-time alert triage [55]. Additionally, BARTPREDICT combines a BART-based predictor with time-series embeddings to anticipate zero-day exploits 24 hours in advance across IIoT power grids [54]. These developments indicate a dual-use trajectory where enhanced generative capabilities simultaneously facilitate both system protection and exploitation. The authors in [53] propose an IoT cybersecurity framework combining LLMs with LSTM networks. The authors in [169] investigate security challenges between IoT devices and LLMs, focusing on adversarial attacks against Llama-2-7b. Their experiments achieve 76% ASR, bypassing alignment measures through prompt injection and gradient-guided search methods.

5.2 Satellite Networks

LLM-based agents could attempt to spoof or manipulate the unencrypted parts of satellite communications. Hassanin *et al.* have developed PLLM-CS, a domain-specific LLM that analyzes satellite telemetry and identifies kinetic-level anomalies in Low-Earth-Orbit constellations [85]. Agnew *et al.* demonstrate that integrating an LLM with a software-defined network controller enables preemptive detection of zero-day routing attacks in LEO mega-constellations through network metric prediction, achieving a 42% reduction in mean detection time [5]. While these implementations position LLMs as defensive tools, they also reveal the potential for adapting these capabilities for satellite-borne intrusions.

5.3 Mobile Ad-Hoc Networks

In Mobile Ad-Hoc Networks (MANETs), where there is no fixed infrastructure, a common threat is Sybil attacks or rogue nodes [129]. LLM-based agents can rapidly create or control multiple nodes to disrupt routing or eavesdrop. Mohandas *et al.* implement a compact transformer for routing anomaly classification in vehicular MANETs, demonstrating superior performance of LLM embeddings over traditional features in high-mobility scenarios [129]. Al-Rubaye and Turkben implement generative replay techniques to maintain lightweight LLM detection accuracy despite concept drift, advancing *continual* adversarial adaptation [12]. Notably, Addula *et al.* present a generative AI-enhanced IDS combining an LLM planner with reinforcement learning, achieving 97% neutralization of multi-vector attacks while generating adversarial traffic for network stress testing [3]. These developments indicate significant potential for autonomous red-teaming.

5.4 Vehicular Networks

Vehicular networks present unique challenges, combining critical latency requirements with extensive and heterogeneous attack surfaces. Therefore, LLM-based agents might exploit these channels for SYN flood DDoS or spoofing attacks. Sun *et al.* in [179] demonstrate that GenAI-driven detection systems effectively analyze vehicular CAN traffic and edge-compute logs, achieving 4.3 percentage points higher recall than CNN baselines in identifying SYN-flood and GPS-spoofing attacks. Begum *et al.* demonstrate LLM capabilities in creating sensor-spoofing payloads that effectively compromise LiDAR-based ADAS, achieving 82% success in triggering emergency braking within a 6G-V2X testbed [30].

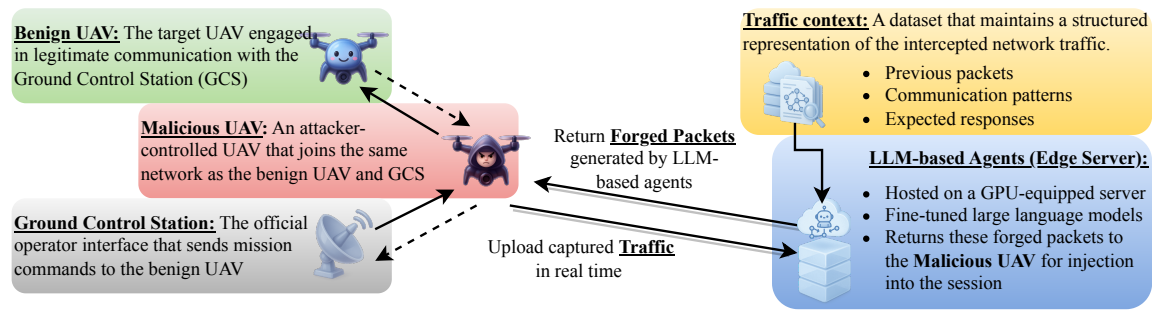


Fig. 7. LLM-based agents for main-in-the-middle attacks with UAV command-and-control.

Shafique *et al.* and Haddaji *et al.* analyze ML countermeasures, noting that prompt-injected LLMs generate polymorphic malware at rates exceeding rule-based gateway blacklisting capabilities [81, 168]. Rajapaksha’s analysis of in-vehicle IDSs highlights risks from fine-tuned GPT agents in automated CAN fuzzing [156]. Aldhyani provides further evidence of deep-learning attack effectiveness against autonomous-vehicle perception systems [15]. These developments exemplify the ongoing competition between LLM-powered offensive and defensive capabilities. The authors in [186] present a comprehensive study on using LLMs for automotive cybersecurity research, developing a customized model called HackerGPT that generates exploitation scripts targeting vehicle systems.

5.5 UAV Networks

As shown in Fig. 7, UAV networks face cyber and kinetic risks through LLM-driven man-in-the-middle attacks. A malicious UAV inserts itself between a ground-control station and a benign UAV to capture TCP packets. An edge server stores traffic and uses LLM agents to predict legitimate packet fields [151]. The edge server returns these forged packet templates to the malicious UAV, which then injects them back toward either the GCS or the benign UAV while optionally suppressing real packets. By repeating this capture-predict-inject loop in real time, the attacker can seamlessly impersonate either party, modify commands, and exfiltrate data without disrupting the appearance of normal communications. Common cyberattacks in UAV networks include GPS spoofing, C2 hijacking, jamming of communication links, and sensor data manipulation. Sedjelmaci *et al.* addresses routing misbehavior through Bayesian learning [166]. Current capabilities now extend to AI-automated spoofing, hijacking, and jamming tactics, as systematically surveyed by Kong [106]. LLM-based agents significantly amplify these threats by autonomously generating attack scripts [165]. Recent studies further highlight dual-use risks, as Dahiya *et al.* in [48] and Garg *et al.* in [46] demonstrate LLM capabilities in generating precise flight control modification scripts.

5.6 Underwater Networks

Underwater networks face unique challenges with bandwidth and latency constraints that were once thought to provide security benefits. These networks are actually susceptible to various security threats, including DoS attacks, spoofing, jamming, and routing attacks. LLM-based agents can now autonomously exploit these vulnerabilities through sophisticated techniques like adaptive DoS floods and automated topology inference. Altameemi *et al.* in [20] demonstrate enhanced anomaly detection capabilities through an SVM-RNN architecture augmented with GPT-generated features, achieving 96.4% accuracy in challenging channel conditions [20]. They identify denial-of-service vulnerabilities susceptible to LLM

Table 8. Representative LLM-based agent cyberattacks on infrastructure-free networks.

Ref.	Agent Architecture	Network Type	Attack Goal	Blue-team Impact
[50, 100, 183]	Multi-agent CoT & ReAct planner	Social Networks	Disrupt decision-making via misinformation flooding	Trust scoring, identity verification, and anomaly detection required
[119, 152, 181]	Prompt-driven traffic shaping with adaptive evasion	Content Delivery Networks	Saturate edge caches and degrade cache-hit ratio	Real-time provenance validation and adaptive rate-limiting needed
[17, 101, 199]	Code-aware retrieval & static analysis loops	Blockchain	Inject malicious smart contracts and poison consensus models	Fine-grained auditing, anomaly scoring, and peer reputation
[26, 109, 220]	KG memory & reflexive telemetry generation	Digital Twin	Inject deceptive sensor data and modify PLC state safely	Requires runtime certification and reasoning-path explainability
[34, 82, 205]	Multimodal RAG & ReInteract dialogue engine	Immersive XR/VR	Personalized social engineering through affect-aware overlays	Adaptive behavior detection and multimodal trust feedback needed
[50, 100, 146, 191]	Swarm RL with self-reflective memory	Agent Networks	Spread prompt-level misinformation and reduce task success	Memory isolation, prompt sanitization, and agent provenance tracking

exploitation, particularly regarding propagation delays and authentication weaknesses. Jocil *et al.* in [99] demonstrate ChatGPT applications in security toolkit development, while Adam *et al.* in [2] address dataset limitations and advocate for generative model applications in cryptographic testing.

5.7 Lessons Learned for Blue Teams

- (1) **Edge-native Security:** For IoT environments, security controls should be pushed to edge devices like gateways and MEC servers. This includes implementing anomaly detection systems for LLM-based cyberattack agents at network entry points to catch coordinated attacks from LLM-orchestrated threats.
- (2) **Multi-Layer Defense Strategy:** Mobile networks need multiple layers of protection to handle cyber threats from LLM-based agents. For example, in MANETs, this means combining radio monitoring, packet inspection, and host-based protection to quickly catch evolving attack tactics. Similarly, in vehicle networks, critical systems should be segregated with rigorous security checks between layers.

6 Cyberattack Capabilities of LLMs-based Agents on Infrastructure-free Networks

Table 8 outlines representative LLM-agent attack strategies across infrastructure-free networks, highlighting their architectures, network targets, attack goals, and implications for blue-team defense.

6.1 Social Networks

In social networks, LLM-based agents can create and manage fake personas at scale, which can flood social platforms with propaganda, phishing, or manipulative content [201]. For instance, CheatAgent shows that by impersonating recommender-system users, an LLM can steer ranking outcomes and exfiltrate private preference data without tripping anomaly detectors [137]. Earlier work on social-network honeypots demonstrated large-scale, automated creation and curation of fake identities to lure threat actors [143]. When combined with generative text models, such bots now produce

spear-phishing content that is statistically indistinguishable from human prose [63]. This may include analyzing behavior over time for human-like inconsistencies, using graph analysis to spot botnets.

6.2 Content-Delivery Networks

Content-delivery networks (CDNs) and information-centric overlays are vulnerable to several types of attacks [133], including cache saturation (Partition DoS), cache-miss amplification, content poisoning, and forwarding loop creation. Since LLM-based agents can generate large volumes of fake or poisoned content to store in caches, implement content verification where possible. Takashima *et al.* in [181] show that LLM-based agents coordinating many low-rate clients can bypass traditional volumetric, DoS thresholds and still saturate edge caches (partition DoS). Liu *et al.* in [119] highlight how intelligent request shaping maximises cache-miss penalties, pushing excessive origin traffic. Models for availability assessment [152] predict that a mere 3-5% decrease in cache-hit ratio can trigger SLA violations network-wide. When detecting the activities of LLM-based agents, CDNs can activate defenses such as serving stale content to suspected nodes, challenging them with CAPTCHAs, or temporarily isolating those requests.

6.3 Blockchain Networks

LLMs can rapidly identify and exploit vulnerabilities in smart contracts. Xiao *et al.* in [199] demonstrate an autonomous agent that locates re-entrancy and integer-overflow patterns, then patches malicious logic stubs into otherwise legitimate Solidity code, producing “smart-contract malware” with nearly zero human effort. A complementary survey work [17] catalogues GPT-powered phishing kits that fabricate token-airdrop sites and wallet-connect dialogs en masse. To compound the risk, collaborative-learning approaches for blockchain anomaly detection [101] make themselves be poisoned through subtle gradient perturbations introduced by a malicious LLM peer, causing selective blindness to the attacker’s transactions. These observations suggest that defending against LLM-based threats requires not only traditional vulnerability patching but also a deep understanding of agents’ capabilities in reasoning, tool orchestration, and stealthy adaptation.

6.4 Digital Twin Networks

Digital Twins rely on accurate data mirroring physical systems. Therefore, LLM-based cyberattack agents can inject deceptive telemetry or alter the twin’s state to mislead operators. Zheng *et al.* in [220] highlight how injecting deceptive telemetry via an LLM-based agent can mislead predictive-maintenance models, triggering premature or unsafe actuator commands. High-fidelity industrial twins are equally vulnerable: Balta *et al.* in [26] report that a twin-resident agent, when compromised, manipulated PLC set-points while maintaining plausible sensor traces. Aviation studies confirm that prompt-level attacks on twin-embedded copilots bypass traditional air-gap assumptions [109]. Krishnaveni *et al.* in [108] propose an intelligent defense framework that deploys counter-agent honeypots and trust scoring, but stresses the need for runtime certification of LLM reasoning paths.

6.5 Immersive Networks

Augmented/virtual reality (AR/VR) platforms present new attack vectors, such as malicious 3D content or overlay attacks [172]. In particular, LLM-based agents amplify these risks by autonomously generating dynamic, personalized attacks. Happa *et al.* [82] are the first to map extended reality (XR)-specific threats; recent work shows LLM-driven avatars dynamically adapt dialogue tone and visual cues to victims’ affective states [205]. Kilger *et al.* in [102] demonstrate detection of camera spoofing in Mixed Reality, yet admit failure against sophisticated, AI-generated overlays. Malicious